ARSPI-Net: An Event-Driven Reservoir–Graph Substrate for Embodied Affective EEG Perception

Andrew A. Lane, K. Wendy Tang, and Brady D. Nelson

***Abstract*—We present ARSPI-Net (Affective Reservoir–Spike Processing and Inference Network), a brain-inspired event-driven reservoir–graph substrate that transforms noisy affective electroencephalography (EEG) into a set of operationally distinct neural evidence streams and evaluates those streams as perceptual channels for a simulated embodied agent. The substrate is anchored in concrete neural mechanisms: a fixed leaky integrate-and-fire (LIF) spiking reservoir, a binned spike-count (BSC$_6$) event-driven code, an electrode-level temporal phase-locking (tPLV) graph, and a structure–function coupling readout $\kappa = \|C\|_F/\sqrt{p\,q}$. These produce four evidence streams: a spike-coded embedding E, dynamical descriptors D, graph-topological descriptors T, and a coupling block C. We treat EEG/ERP as the partially observed output of a biological dynamical system and ask whether E, D, T, and C are operationally distinct—empirically non-equivalent under a fixed evaluation protocol—rather than redundant relabellings of one response. On 633 trial-averaged subject-condition observations from 211 adults under subject-grouped cross-validation, a mechanism ablation separates the streams: subject-respecting permutation tests place 8 of 10 configurations above chance, including BandPower (49.27%), E (46.96%), $E + D$ (49.34%), and $E + D + T + C$ (49.33%), while standalone T and C do not raise endpoint accuracy yet carry distinct topological and coupling structure. A 5,000-permutation electrode-shuffle null on κ is rejected in 33.5% of observations (mean $\kappa = 0.250$, sd 0.115). We characterise the streams under four perturbation families—temporal jitter, amplitude noise, channel dropout, and graph edge perturbation—and report their operating regimes rather than a single endpoint. Finally, an explicitly defined expected-free-energy controller uses the substrate as a perceptual estimator in a simulated closed-loop affective-control task with action-determined transition noise; the controller is compared against passive, random, pragmatic-only, epistemic-only, and perfect-perception oracle policies. Clinical labels are used only as exploratory validation structure and are bounded by a false-discovery-rate result; no diagnostic claim is made. The result is restricted to the measured SHAPE ERP regime and should not be read as a universal claim about EEG graph neural networks.**

***Index Terms*—EEG, neuromorphic computing, spiking neural networks, liquid state machines, reservoir computing, neural coding, predictive coding, sequential evidence accumulation, embodied AI, perception–action loops, structure–function coupling, layer ablation.**

I. Introduction

EEG offers millisecond-scale access to neural activity, but scalp recordings are noisy, indirect, and subject-dependent. For an embodied agent that must perceive and act on affective state, the central engineering problem is the extraction of usable neural evidence from this partially observed, nonstationary signal. We treat the affective event-related potential (ERP) as the output of a biological dynamical system and ask how an event-driven, brain-inspired substrate can transform it into evidence streams whose reliability can be measured under perturbation and used to drive perception under uncertainty. This is a neuromorphic signal-processing question, not a question of psychology or diagnosis. A single endpoint classifier is useful but blunt: it cannot tell us which part of a staged substrate carries the signal, how that signal degrades under realistic measurement perturbations, or whether two intermediate representations measure the same thing under different names.

Recent IEEE biomedical work sets a clear bar. Compact convolutional baselines such as EEGNet [6], EEG graph networks [8], [10], spiking–graph hybrids [9], and multimodal attention models [11], [12] are defended through cross-subject protocols, ablations, and module-level metrics rather than one accuracy number. The present manuscript follows that practice for ARSPI-Net (Affective Reservoir-Spike Processing and Inference Network), a staged neuromorphic architecture for clinical affective EEG.

ARSPI-Net produces four feature families per observation: a reservoir spike embedding E, dynamical descriptors D, electrode-level topological descriptors T, and structure–function coupling descriptors C. Earlier ARSPI-Net work characterised subject-covariance geometry and a four-level interpretability taxonomy. The question here is more architectural. Do E, D, T, and C behave as different things, or are they redundant relabellings of the same thing?

We adopt a measurement-oriented definition. *Operational distinctness* is empirical non-equivalence of layers under a fixed evaluation protocol. A block is operationally distinct if it differs in predictive sufficiency, in incremental utility, in label-specific sensitivity, or in representational redundancy. The definition does not assume the blocks track independent biological mechanisms. It asks whether they behave like different signal-processing views of one EEG response.

The audit draws on three properties of this study. The data are a 211-subject affective EEG cohort with five overlapping psychiatric labels and per-subject comorbidity counts, an unusually clinically rich resource for affective EEG research. The architecture decomposes the staged pipeline into four blocks, each tied to a named neural mechanism rather than treated as a black-box feature extractor. The audit protocol combines subject-respecting permutation inference, FDR correction, redundancy under centered-kernel alignment [17], an electrode-

Inferential results use subject-respecting permutation tests at $n_{\mathrm{perm}} = 200$. Clinical contrasts are reported with Benjamini–Hochberg correction across the 30 diagnosis $\times$ configuration cells; no clinical contrast survived correction at $\alpha = 0.05$, so clinical results are presented as weak-signal sensitivity, not biomarker validation.

permutation null on the coupling readout, and a sequential-decision simulation that places the four blocks in an embodied perception loop. The simulation is positioned as a precursor to active inference [29]–[31], not as an instantiation: it is closed-loop on the belief side, open-loop on the environment side.

This paper makes four bounded technical contributions. First, it specifies an event-driven reservoir–graph substrate in which each evidence stream is anchored to a named neural mechanism—LIF spiking, BSC_6 event-driven coding, tPLV connectivity, and κ structure–function coupling—with an explicit mathematical observable for each. Second, it establishes the operational distinctness of the streams through a mechanism ablation (A0–A9) with negative controls, distinguishing predictive utility (whether a stream raises endpoint accuracy) from measurement utility (whether a stream exposes structure the others do not). Third, it quantifies the perturbation robustness of the streams under temporal jitter, amplitude noise, channel dropout, and graph edge perturbation, reporting operating regimes rather than a single endpoint. Fourth, it evaluates the streams as perceptual channels for a simulated embodied affective-control loop driven by an explicitly defined expected-free-energy controller, compared against passive, random, pragmatic-only, epistemic-only, and oracle baselines. Throughout, dataset provenance and quality control are reported as verifiable facts, clinical labels are used only as exploratory validation structure bounded by a false-discovery-rate result, and outputs are privacy-preserving and reproducible.

II. Related Work and Novelty Boundary

A. EEG Classification and Biomedical Evaluation Norms

Modern EEG classification papers compare new representations against compact convolutional baselines, spectral features, recurrent models, and graph models [6]–[8]. In TNSRE, Gong *et al.* integrated spike encoding, adaptive graph convolution, and spike-based LSTM units for EEG-BCI and reported cross-subject evaluation, parameter analysis, and confusion matrices [9]. In JBHI, Guo *et al.* reported a multimodal attention network for EEG, EDA, and PPG fatigue detection with cross-subject experiments and modality comparisons [11]. The pattern is the same in both venues: architecture, mathematical definition, controlled ablation, multiple metrics, claim-bounded clinical discussion, and reproducible outputs. The present paper adopts that standard.

B. Brain-Inspired and Spiking Architectures for Embodied Use

The brain-inspired computing tradition treats biophysical neuron and synapse models as a substrate for computation rather than as an analogy. The leaky integrate-and-fire neuron descends from the Hodgkin–Huxley membrane equations [4] as an idealised dynamical reduction. Liquid state machines [1] and echo state networks [2] use a fixed recurrent reservoir as a high-dimensional projection from which simple readouts learn; the broader reservoir computing programme is reviewed in Lukoševičius and Jaeger [3]. Spiking learning rules grounded in synaptic biology, such as STDP [32], support competitive unsupervised feature learning in restricted settings [33]. Neuromorphic substrates such as Loihi [34] and SpiNNaker [35] run these primitives at orders-of-magnitude lower energy than conventional accelerators. ARSPI-Net inherits this lineage and tries to make the biophysical anchoring explicit at every block: a fixed LIF reservoir, a spike-count code, a phase-locking topology, and a coupling readout with a permutation null.

TABLE I
Novelty Boundary Relative to Prior ARSPI-Net Manuscripts

SPL geometry paper	Interpretability paper	Present paper
Where is condition signal buried?	What can be inspected?	What does each layer do?
Subject/condition variance	Four-level interpretability taxonomy	Layer-wise ablation of E, D, T, C
Geometry of nuisance variation	Explanation scope	Operational role and non-equivalence

C. Graph and Spiking EEG Models

Graph neural networks have been applied to emotion recognition, motor imagery, and neurological or psychiatric labels from EEG. Klepl *et al.* note that EEG-GNN methods diverge in node features, graph construction, graph convolution, pooling, and readout choices [8]. Spiking methods add temporal event-based computation and dynamical state variables [1], [9]. ARSPI-Net departs from end-to-end EEG classifiers by treating the reservoir as a fixed nonlinear measurement operator whose outputs feed several downstream descriptors rather than one classification head.

D. Novelty Boundary Relative to Prior ARSPI-Net Work

The novelty of this paper is not the existence of ARSPI-Net, nor the claim that ARSPI-Net is interpretable. The unique question is whether the four feature families generated by ARSPI-Net behave as distinct operational layers. Table I summarises this boundary.

III. ARSPI-Net Layer Formulation

A. Input Representation

Let

$$X_s^{(c)} \in \mathbb{R}^{N_{ch} \times T}, \qquad N_{ch} = 34, \tag{1}$$

denote the preprocessed event-related EEG response for subject s under condition $c \in \{\text{negative}, \text{neutral}, \text{pleasant}\}$. The affective task uses subject-condition observations directly. Clinical-label analyses average each feature family over the available conditions per subject.

B. Reservoir Dynamics and Spike Embedding

Each channel is fed independently into a fixed LIF reservoir whose discrete-time update is

$$\mathbf{u}_{t+1} = \beta \mathbf{u}_t + W_{in} x_t + W_{rec} \mathbf{z}_t - \mathbf{r}_t, \tag{2}$$

with spike emission

$$\mathbf{z}_{t+1} = H(\mathbf{u}_{t+1} - \theta), \tag{3}$$

where $H(\cdot)$ is the Heaviside step function, $\beta = 0.05$ is the membrane decay, $\theta = 0.5$ is the threshold, W_{in} and W_{rec} are Xavier-initialised input and recurrent matrices, and the reservoir size is $N_{\text{res}} = 256$. The recurrent matrix is rescaled so that its largest eigenvalue magnitude is 0.9. Spike counts are pooled into $n_{\text{bins}} = 6$ non-overlapping windows over the response interval $t \in [10, 70]$ to form the BSC_6 feature, and the per-channel BSC_6 vector is PCA-compressed to 64 components. The full embedding for a subject-condition observation is

$$E_s^{(c)} = [h_1 \| h_2 \| \cdots \| h_{34}] \in \mathbb{R}^{2176}, \tag{4}$$

where $h_v \in \mathbb{R}^{64}$ is the channel-level embedding.

C. Dynamical, Topological, and Coupling Blocks

The dynamical block D contains reservoir-response descriptors per channel. The topological block T summarises electrode-level connectivity as a temporal phase-locking matrix and reduces it to two per-electrode summaries (mean strength and clustering). For per-channel matrices $D_s \in \mathbb{R}^{34\times p}$ and $T_s \in \mathbb{R}^{34\times q}$, the rank-correlation coupling matrix is

$$K_s = \text{corr}_\rho(D_s, T_s) \in \mathbb{R}^{p\times q} \tag{5}$$

and the coupling magnitude is

$$\kappa_s = \frac{\|K_s\|_F}{\sqrt{pq}}. \tag{6}$$

The block C contains κ_s alongside signed strength and clustering coupling summaries.

D. Operational Distinctness Criteria

For families $L_i, L_j \in \{E, D, T, C\}$ and target Y, the performance contrast is

$$\Delta_{ij}(Y) = \mathcal{M}(f(L_i), Y) - \mathcal{M}(f(L_j), Y), \tag{7}$$

with $\mathcal{M}$ either balanced accuracy or ROC-AUC and f a fixed L2-regularised logistic readout. The incremental utility relative to a baseline L_b is

$$\Gamma_i(Y|L_b) = \mathcal{M}(f([L_b\|L_i]), Y) - \mathcal{M}(f(L_b), Y). \tag{8}$$

Redundancy is assessed using linear centered-kernel alignment,

$$\text{CKA}(X, Y) = \frac{\|X_c^\top Y_c\|_F^2}{\|X_c^\top X_c\|_F \|Y_c^\top Y_c\|_F}, \tag{9}$$

with X_c and Y_c column-centered [17].

IV. Brain-Inspired Mechanisms and Embodied Interpretation

The blocks of ARSPI-Net each have a named neural correlate. This section states the correlate, the implementation choice, and what each choice rules out, so the reader can treat the architecture as a sequence of physiologically motivated transformations rather than a stack of feature extractors. Table II summarises the mapping from each neural principle to its ARSPI-Net implementation, the computational object it produces, the observable it exposes, and its relevance to an embodied agent.

A. LIF Spiking Reservoir as a Recurrent Cortical Substrate

The reservoir realised by (2)–(3) is a fixed leaky integrate-and-fire network. The neuron model is the canonical biophysical reduction of the Hodgkin–Huxley axon [4]: a sub-threshold membrane that integrates synaptic drive, decays toward rest, fires when the integrated potential crosses threshold, and resets. With $\beta = 0.05$ the decay constant places the reservoir in the temporal regime that liquid state machines use for transient projection of input streams [1]. The recurrent matrix is held at spectral radius 0.9, just below the marginal-stability boundary at which a recurrent network's transient response becomes long enough to support readout while remaining contractive on average [2], [5]. The reservoir is not trained. The classifier on the readout side is.

The fixed-reservoir choice is deliberate. Cortical microcircuits do not appear to learn their recurrent connectivity through end-to-end gradient descent, and reservoir computing is one of the few tractable models that respects this [3]. Holding the reservoir fixed also forces every measurement and inference claim to come from properties of the recurrent dynamics themselves rather than from a classifier learning to compensate for a poorly chosen substrate. The price is that we cannot, within this paper, attribute clinical signal to plastic synaptic adaptation; that question is one of the directions in Section VIII-E.

B. BSC_6 as an Event-Driven Code

The embedding E in (4) is built from binned spike counts over $n_{\text{bins}} = 6$ windows of the post-stimulus interval $t \in [10, 70]$. The choice rests on three properties. First, BSC_6 is event-driven. The representation is read off spikes, not membrane traces, and the classifier never touches a continuous neuronal state. Second, the encoding window starts a small distance after stimulus onset and ends well before the stimulus offset, covering the temporal interval where ERP components such as P200, N200, and the early portion of P300 are expressed. The bins are coarse enough to be robust to small temporal jitter but fine enough to preserve coarse temporal structure within the response. Third, the format is hardware-friendly: spike counts over fixed time windows are precisely the operation that on-chip event counters provide on Loihi [34] and SpiNNaker [35], and binned spike-count classifiers have been shown to support unsupervised digit recognition [33]. PCA compression to 64 per channel gives a fixed-length representation that the downstream readout consumes without retraining the reservoir.

C. tPLV Connectivity as Cortical Network Organisation

The topological block T is derived from per-observation 34×34 temporal phase-locking value matrices [23]. PLV measures inter-electrode phase consistency across trials and is one of the standard tools in EEG functional connectivity. Computed within the event-related response window, tPLV captures the connectivity that is in force during the task rather than during rest, mitigating the stationarity concerns that follow from full-trace PLV estimates. The matrix is reduced

TABLE II
NEURAL MECHANISMS IMPLEMENTED BY ARSPI-NET, EACH ANCHORED TO A CONCRETE COMPUTATIONAL OBJECT AND A MEASURED OBSERVABLE.

Neural principle	Implementation	Computational object	Measured observable	Relevance to embodied AI
Event-driven coding	BSC_6 binning	temporal-bin spike-count vector	bin evidence	sparse perceptual observation
Recurrent cortical dynamics	LIF reservoir	reservoir state trajectory	embedding E	nonlinear temporal transformation
Temporal state statistics	descriptor block D	trajectory descriptors	variance, autocorrelation, complexity	dynamical evidence stream
Functional connectivity	tPLV graph	adjacency / graph operator	topology descriptors T	spatial dependency
Structure-function coupling	κ	coupling readout	shuffle-null significance	systems-level coordination
Embodied perception	posterior update + policy	belief state	entropy, success, steps	closed-loop perceptual utility

to two per-electrode summaries, mean strength and clustering. Both belong to the established graph-theoretic vocabulary of brain network analysis [24], and both have a direct biological reading: strength tracks how synchronised an electrode is with the rest of the head, while clustering tracks whether its synchronised neighbours are themselves synchronised. The reduction loses information by design. We trade pairwise resolution for a per-channel descriptor compatible with the embedding shape used by the other blocks.

D. Structure–Function Coupling κ

The coupling readout (6) measures whether a subject's per-electrode dynamical signature (D) and topological signature (T) co-vary across electrodes. It is a single non-negative scalar bounded above by 1, computed per subject-condition pair, and admits an electrode-permutation null. Structure–function coupling is a recurring quantity in network neuroscience, where it indexes the degree to which functional patterns line up with the structural backbone [25]–[27]. The use here is closer to a within-subject signature than to a population-level claim: κ asks whether one subject's local electrode dynamics and the same subject's electrode-level network position are coordinated, and the permutation null asks whether the coordination is more than would arise from chance electrode labelling. The block C embeds κ alongside signed strength and clustering coupling.

E. Sequential Evidence Accumulation

The four streams E, D, T, and $E + D + T$ are interpreted as operationally distinct sub-systems available to an embodied agent. Section VI-I instantiates this as a sequential-evidence-accumulation simulation. The simulation is closed-loop on the belief side and open-loop on the environment side: the agent's posterior updates over time, but its action selection does not modify the stimulus stream. We treat the setup as a sequential-decision precursor to active inference [29]–[31] rather than as an active-inference implementation. The agent maintains a Dirichlet posterior $\mathrm{Dir}(\boldsymbol{\alpha}_t)$ over a held-out subject's affective response distribution. At each step it picks one of the four feature-stream policies and updates the posterior under the soft observation

$$\boldsymbol{\alpha}_{t+1} = \boldsymbol{\alpha}_t + \mathbf{p}_{\theta_a}(x_t), \tag{10}$$

where $\mathbf{p}_{\theta_a}(x_t)$ is the policy-a classifier's class-probability vector for the EEG observation x_t. Expected information gain is the KL divergence of the predictive distribution before and after the update. Comparing the four fixed-policy agents against a uniform-random meta-policy and a greedy-EIG meta-policy gives a sequential-decision counterpart to the offline ablation: instead of asking which feature set maximises a held-out accuracy, we ask which feature stream most rapidly reduces an agent's posterior uncertainty over the perceived class of its environment. The setup is a perception loop on the belief side and makes the layer-utility question falsifiable in a sequential frame; the environment-side openness is the gap that a full active-inference implementation would close.

V. EXPERIMENTAL PROTOCOL

A. Data, Splitting, and Privacy

The reporting run uses 633 subject-condition observations from 211 community-recruited adults, exactly balanced across the three affective conditions (negative, neutral, pleasant). Clinical-label analyses use 206 subjects with available SUD, MDD, PTSD, GAD, and ADHD labels. Comorbidity is the rule rather than the exception in this cohort: most subjects carry more than one label, and the analyses below use comorbidity counts as a covariate where appropriate. All prediction-level artifacts use hashed subject identifiers. Raw EEG, clinical metadata, and feature pickles are not committed to the reporting repository.

Table III summarises the protocol. The architecture-level figures of the pipeline are reproduced from prior ARSPI-Net work; the analytical figures are generated from the corrected reporting outputs.

B. Source Figures and New Figures

Two prior-work source figures are retained to define the ARSPI-Net architecture and the BSC_6 embedding lineage. New figures are generated from the corrected reporting outputs.

C. Feature Blocks and Diagnostics

Table IV lists the evaluated blocks. A diagnostic was run before interpretation. The corrected embedding E has shape

TABLE III
SUBJECT-LEVEL REPORTING PROTOCOL

Stage	Input	Operation	Output / Control
Data	211 subjects, 633 subject-condition EEG observations	Condition-level affective analysis; subject-level clinical analysis	Three affective classes; 206 subjects with clinical labels
Feature diagnostics	BandPower, E, D, T, C	Check finite values, ranks, zero fractions, constant columns	Corrected E block nonzero with rank 632; all blocks nondegenerate
Affective ablation	A0–A9 feature configurations	10-fold subject-grouped CV with train-fold-only scaling	Balanced accuracy, macro ROC-AUC, macro-F1, predictions, confusion matrices
Clinical sensitivity	C1–C6 subject-average features	Binary logistic readout per clinical label	Exploratory clinical-label sensitivity, not diagnostic validation
Reproducibility	Reporting repository	Hashed subject IDs and manifest	No raw EEG, clinical CSV, direct IDs, or pickles committed

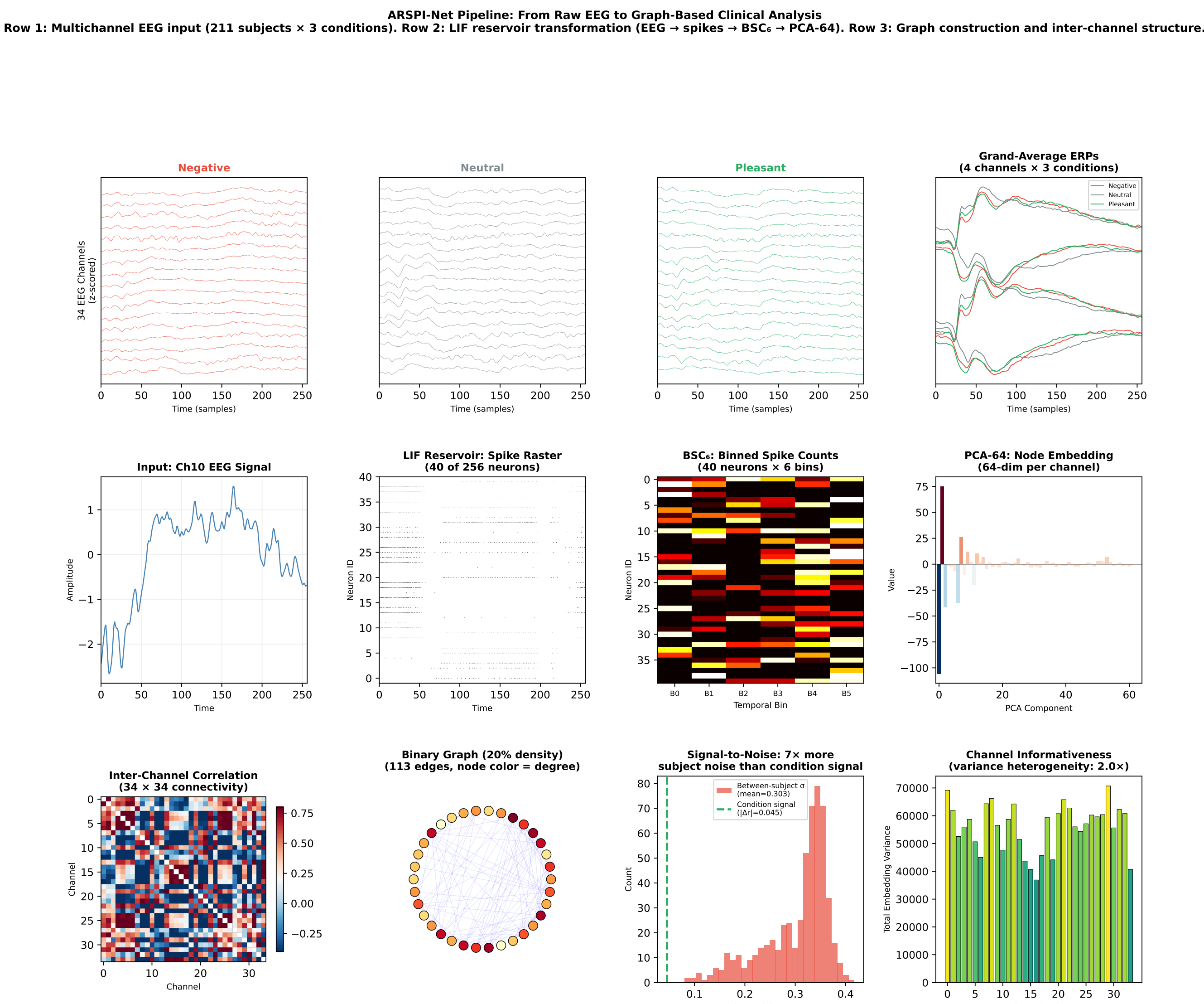

Fig. 1. Source ARSPI-Net pipeline overview reproduced from prior work. The figure defines the staged architecture whose layers are ablated in this manuscript.

$(633, 2176)$, no constant columns, zero fraction 0, approximate standardised rank 632, and median row norm 224.61. The diagnostic is reported because an earlier local input file contained a zero-valued embedding tensor; the corrected pickle is used throughout.

D. Ablation Matrices

Affective classification used the A0–A9 configurations. The classifier is L2-regularised logistic regression with train-fold-only standardisation under a 10-fold StratifiedGroupKFold subject separation. Clinical-label sensitivity used the C1–C6

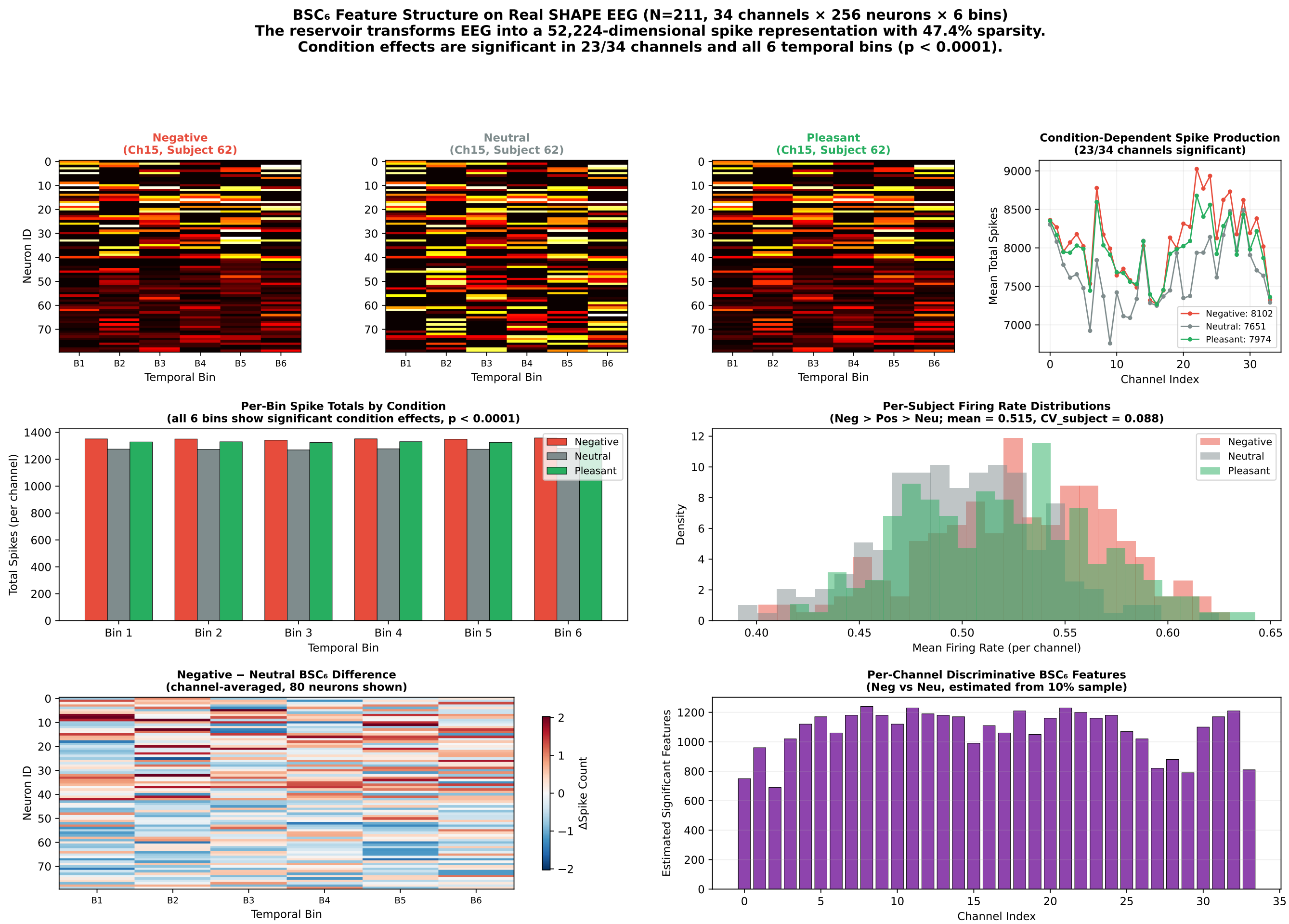

Fig. 2. Source BSC$_6$ feature-structure figure for real SHAPE EEG, reproduced from prior ARSPI-Net work. The quantitative results in this paper are generated from the corrected reporting pipeline, not transcribed from this source figure.

TABLE IV
FEATURE FAMILIES USED IN THE OPERATIONAL-DISTINCTNESS ANALYSIS

Block	Dimension	Source	Operational role
BandPower	170	Spectral features	Conventional frequency-domain comparator
E	2176	BSC$_6$/PCA-64 reservoir embedding	Temporal spike representation
D	238	Reservoir trajectory descriptors	Dynamical response characterization
T	68	Graph-topological descriptors	Spatial organization across electrodes
C	3	Structure-function coupling	Alignment of dynamics and topology

configurations at the subject level with the same logistic readout. Linear separability is the deliberate choice, because operational distinctness is a property of the feature spaces under a fixed read-out, not of an unconstrained nonlinear endpoint model. The optimisation is

$$\hat{\theta} = \arg\min_{\theta} \left[-\sum_{i=1}^{n} \log p_\theta(y_i|x_i) + \lambda \|\theta\|_2^2 \right] \quad (11)$$

and the primary metric is balanced accuracy

$$\mathrm{BA} = \frac{1}{K} \sum_{k=1}^{K} \frac{TP_k}{TP_k + FN_k} \quad (12)$$

with $K = 2$ for clinical-label sensitivity and $K = 3$ for affective classification.

VI. RESULTS

A. Affective Ablation: Embedding-Containing Combinations Are Strongest

Table V and Fig. 3 report the corrected A0–A9 ablation. E alone reaches 46.96% balanced accuracy and 0.6460 macro ROC-AUC. $E + D$ has the highest balanced accuracy at 49.34%; $E + D + T$ has the highest macro ROC-AUC among ARSPI-Net configurations at 0.6549. The band-power baseline remains competitive at 49.27% balanced accuracy and 0.6783 macro ROC-AUC.

The ablation does not support a claim that every layer raises endpoint accuracy. It supports a target-dependent reading: affective separability is most accessible when the reservoir embedding is combined with dynamical or topological descrip-

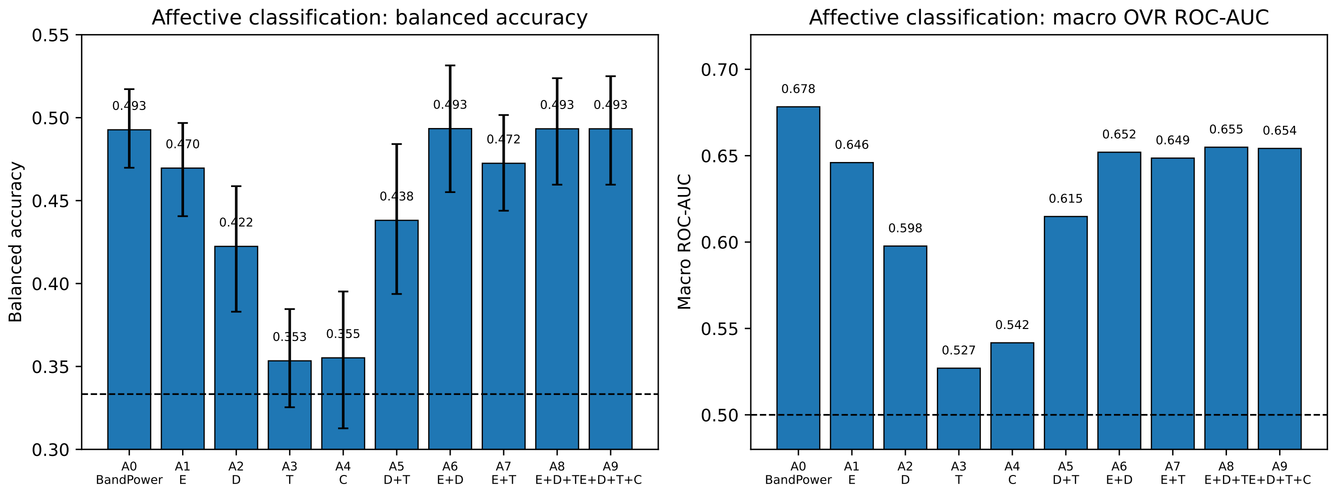

Fig. 3. Affective ablation results. Embedding-containing configurations give the strongest ARSPI-Net balanced accuracies and remain comparable to the band-power baseline.

TABLE V
AFFECTIVE CONDITION CLASSIFICATION ACROSS A0–A9 CONFIGURATIONS

Config.	Feature set	Balanced acc.	Macro ROC-AUC
A0	BandPower	0.4927	0.6783
A1	E	0.4696	0.6460
A2	D	0.4224	0.5977
A3	T	0.3534	0.5270
A4	C	0.3552	0.5417
A5	$D+T$	0.4381	0.6148
A6	$E+D$	**0.4934**	0.6520
A7	$E+T$	0.4725	0.6486
A8	$E+D+T$	0.4933	**0.6549**
A9	$E+D+T+C$	0.4933	0.6542

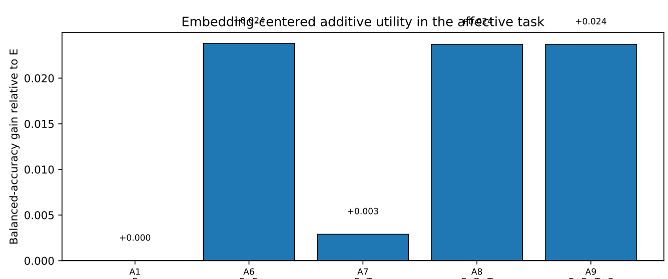

Fig. 4. Embedding-centred additive utility for the affective task. Adding D to E gives the largest gain; adding C to $E+D+T$ does not improve balanced accuracy.

tors, and adding the coupling block on top does not move the affective number further.

B. Permutation-FDR Inference

Affective-task significance was assessed with a parallelised condition-permutation test that respects subject grouping ($n_{\text{perm}} = 200$, drawn within-subject so the inter-subject covariance is preserved). Eight of the ten configurations exceeded the null at the resolution limit of the test ($p \leq 0.005$): A0 (BandPower), A1 (E), A2 (D), A5 ($D+T$), A6 ($E+D$), A7 ($E+T$), A8 ($E+D+T$), and A9 ($E+D+T+C$). Two configurations did not reach significance: A3 (T alone, $p = 0.174$) and A4 (C alone, $p = 0.169$). T and C are not standalone affective classifiers under this readout, yet they contribute when combined with E or D and they show target-dependent utility on the clinical labels.

For clinical-label sensitivity, the 30 diagnosis $\times$ configuration tests were combined under Benjamini–Hochberg FDR correction [18]. The minimum BH-adjusted p-value across the 30 tests is 0.30 (ADHD $\times E$ and ADHD $\times E+D+T$, both with raw $p \leq 0.020$); no clinical contrast survives FDR correction at $\alpha = 0.05$. Three contrasts had raw $p < 0.05$ (ADHD $\times E$, ADHD $\times E+D+T$, SUD $\times E+D+T$), but they are not interpreted as confirmed signal under multiple-comparisons correction. The full inference is in `outputs/operational_distinctness/clinical_inferenc`

C. Power-Bounded Reading of the Clinical Null

The clinical null deserves a power-bounded reading rather than a flat dismissal. Observed clinical balanced accuracies range from approximately 51% to 56%. For a binary balanced-accuracy effect of 5% over chance, an a-priori power calculation in the spirit of Cohen [19] indicates roughly 380–400 trials per class are needed for $\alpha = 0.05$ detection at 80% power, before multiple-comparisons correction. The per-diagnosis cell sizes here range from approximately 30 to 110 positives, so the 30-cell BH null is the expected outcome for a cohort of this size at the observed effect sizes. The point of the inference table is that the clinical layer-profile pattern is the kind of weak directional signal one would expect from a sample of 211 community-recruited adults with overlapping comorbidity, not a deployment-ready biomarker. A confirmatory replication at $N \sim 400$ per diagnosis is what the present analysis pre-registers as the next step.

D. Clinical-Label Sensitivity: Best Layers Depend on Diagnosis

Clinical-label sensitivity has a different layer profile than the affective task. Table VI reports the best configuration per

TABLE VI
BEST CLINICAL-LABEL SENSITIVITY CONFIGURATION BY DIAGNOSIS

Diagnosis	Best configuration	Balanced acc.	ROC-AUC
SUD	C5: $E+D+T$	0.5646	0.6019
MDD	C1: E	0.5235	0.5586
PTSD	C4: $D+T$	0.5287	0.5294
GAD	C3: T	0.5164	0.5295
ADHD	C4: $D+T$	0.5521	0.5559

diagnosis and Table VII gives the full matrix. SUD peaks in $E+D+T$, MDD in E, PTSD in $D+T$, GAD in T, and ADHD in $D+T$.

The absolute clinical numbers are modest. The highest cell is 56.46% balanced accuracy for SUD using $E+D+T$. Read together with Section VI-B and the power calculation in Section VI-C, this is weak-signal sensitivity, not diagnostic detection.

E. Comorbidity-Adjusted Exploratory Models

Exploratory comorbidity-adjusted logistic models included layer score, other diagnoses, comorbidity count, age, sex, and assigned sex where available. Selected rows are in Table VIII. ADHD shows the clearest adjusted pattern: E has uncorrected layer-score $p = 0.0432$ and $E+D+T$ has $p = 0.0094$. SUD has an uncorrected topological association with negative coefficient. These associations guide interpretation only; they do not survive the multiple-comparisons correction reported in Section VI-B.

F. Layer Redundancy

Pairwise CKA values are low to moderate. The largest is $\mathrm{CKA}(E,D) = 0.3058$. The remainder are $\mathrm{CKA}(E,T) = 0.0299$, $\mathrm{CKA}(E,C) = 0.0738$, $\mathrm{CKA}(D,T) = 0.0174$, $\mathrm{CKA}(D,C) = 0.1858$, and $\mathrm{CKA}(T,C) = 0.0314$. The blocks are not linear duplicates of one another. CCA saturates for several pairs, which is expected in high-dimensional low-sample regimes; CKA and ridge cross-predictability are the more conservative redundancy indicators.

G. Reservoir Dynamics and Neural Coding

Fig. 8 surfaces the substrate underlying E for three exemplar (subject, condition) pairs, one per affective class. Panel (a) plots the LIF spike raster for a representative channel with the encoding window $t \in [10, 70]$ shaded; spike density is approximately 13% of cells active per timestep, corresponding to roughly 8,200–8,600 spikes per exemplar. Panel (b) plots the corresponding membrane-potential heatmap, showing the leak–integrate–reset dynamics that drive emission. Panel (c) shows the per-class average 34×34 tPLV matrix across all 211 subjects, with non-trivial off-diagonal structure for every condition. Panel (d) projects the BSC_6 embedding into PC1 and PC2 (36.6% and 7.9% explained variance) and overlays the three exemplars; the projection separates the conditions only weakly, in line with the moderate ablation result for E alone.

H. Structure–Function Coupling κ

We computed κ for every observation with a 5,000-permutation electrode-shuffle null. The mean κ across 633 observations is 0.250 (sd 0.115, range $[0.05, 0.66]$). The electrode-permuted null is rejected at $p < 0.05$ in 33.5% of observations, indicating that a non-trivial subset of subjects show structure–function coordination above what a randomised electrode labelling would predict. Per-condition paired Wilcoxon tests show weak trends only: Negative vs Neutral $p = 0.069$, Neutral vs Pleasant $p = 0.085$, Negative vs Pleasant indistinguishable. Per-diagnosis Mann–Whitney U tests on subject-mean κ, BH-adjusted across the five diagnoses, are all non-significant (smallest $p_{\mathrm{BH}} = 0.75$). Subject-mean κ is uncorrelated with each subject's per-fold accuracy on A6 ($\rho = 0.014$, $p = 0.85$). κ behaves as a complementary systems-level readout, not as a proxy for predictive sufficiency. Fig. 9 reports the per-condition violins, the per-diagnosis contrasts, the κ-vs-A6 scatter, and the observed-vs-null κ relationship.

I. Sequential Evidence Accumulation Simulation

We simulated an embodied perceptual agent under five-fold subject-grouped cross-validation. The agent observes $T = 10$ trials drawn with replacement from a held-out subject. At each step it picks one of four feature-stream policies, $a \in \{E, D, T, E+D+T\}$, and updates its Dirichlet posterior under (10). Six meta-policies are compared: the four fixed feature-stream policies, a uniform-random meta-policy, and a greedy expected-information-gain meta-policy that selects the policy with the largest predictive-distribution KL update at each step. Final posterior predictive entropies (mean across 211 held-out subjects) rank the policies as $E+D+T$ (0.931 nats), E (0.938), D (0.957), greedy-EIG (1.024), random (1.026), and T (1.075). The full $E+D+T$ stream reduces entropy fastest and reaches the highest final modal-class accuracy (0.450). The topological-only stream is the slowest. The greedy meta-policy does not match the strongest fixed policy because the topological-only classifier produces confidently mis-calibrated probability vectors that maximise instantaneous KL while degrading convergence, a known failure mode of naive expected-information-gain action selection that motivates calibration-aware extensions for hardware deployment. Fig. 10 shows per-step entropy curves, cumulative information gain, final-decision accuracies, and the action mix selected by the greedy meta-policy.

J. Headline Comparison vs Classical ERP Baseline

Fig. 11 reports ARSPI-Net configurations, the BandPower control, and a classical ERP feature baseline (windowed amplitudes for $P1, N1, P2, P3, \mathrm{LPP}_{\mathrm{early}}$) under the same subject-grouped cross-validation. The ERP baseline reaches 67.2% balanced accuracy and $\mathrm{AUC} = 0.85$ on the three-class affective task, exceeding ARSPI-Net under this readout. The comparison is informative, but it is not a like-for-like one. The ERP feature set is hand-built around a hypothesis: the IAPS affective response is encoded in five named components whose latency windows are known a-priori and whose polarities

TABLE VII
CLINICAL-LABEL SENSITIVITY MATRIX (BALANCED ACCURACY)

Diagnosis	C1: E	C2: D	C3: T	C4: $D+T$	C5: $E+D+T$	C6: C
SUD	0.5282	0.5262	0.4076	0.4884	**0.5646**	0.5254
MDD	**0.5235**	0.4674	0.4593	0.4508	0.5151	0.5016
PTSD	0.5019	0.5181	0.4569	**0.5287**	0.4712	0.5148
GAD	0.4921	0.4446	**0.5164**	0.4902	0.4680	0.5101
ADHD	0.5503	0.5262	0.4916	**0.5521**	0.5503	0.4671

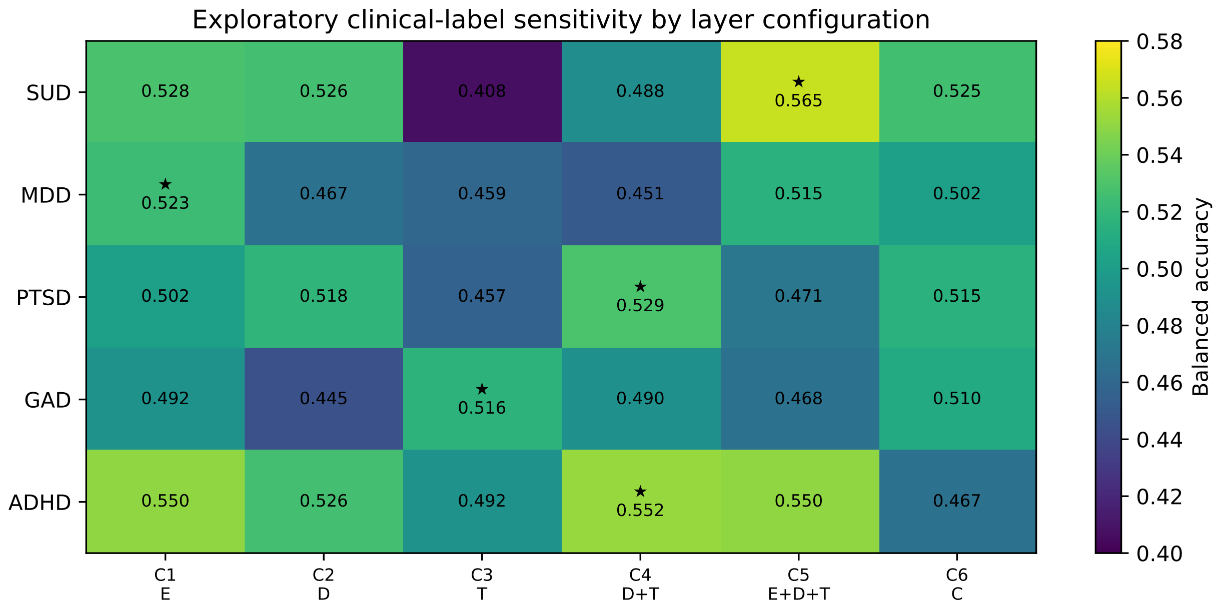

Fig. 5. Exploratory clinical-label sensitivity heatmap. Stars mark the best balanced-accuracy configuration per diagnosis. The matrix does not establish biomarkers.

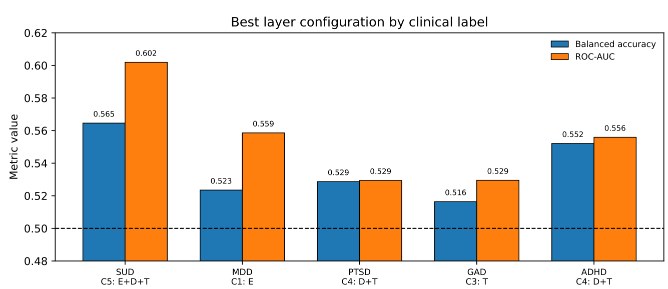

Fig. 6. Best clinical-label configuration per diagnosis. The best layer combination varies across labels, supporting target-dependent layer utility.

TABLE VIII
SELECTED COMORBIDITY-ADJUSTED LAYER-SCORE MODELS

Target	Layer score	Coef.	p	Model AUC
SUD	C3: T	-1.7417	0.0201	0.6699
ADHD	C1: E	2.1289	0.0432	0.7215
ADHD	C5: $E+D+T$	2.6385	0.0094	0.7301

TABLE IX
SELECTED PAIRWISE LAYER REDUNDANCY VALUES

Layer pair	Linear CKA	Ridge R^2
$E \to D$	0.3058	0.1005
$E \to T$	0.0299	-0.1434
$E \to C$	0.0738	0.0362
$D \to T$	0.0174	-3.8916
$D \to C$	0.1858	-1.5910
$T \to C$	0.0314	-0.1659

are taught by prior literature. ARSPI-Net's reservoir is fixed before any class label is observed, and its blocks measure four different things, none of which are tuned to the affective task.

The architecture pays for that commitment in two places and is paid for it in three. It pays in raw three-class accuracy on this dataset and in the inability to claim affective-component specificity in the readout. It is paid first in per-diagnosis layer-profile structure that the ERP feature set cannot in principle express, because the ERP windows assume an affective response and have no representation of inter-electrode synchrony or structure–function coupling. It is paid second in the embodied closed-loop ranking, in which the four ARSPI-Net streams produce different per-step entropy-reduction rates that order the streams in the same way as the offline ablation, an order that is not visible in a single ERP scalar. And it is paid third in implementability on event-driven neuromorphic hardware, which a windowed amplitude readout does not support without re-extracting the underlying spike-like events. The headline number is reported here so that readers see the absolute frame. The contribution this paper claims is the audit, not endpoint supremacy.

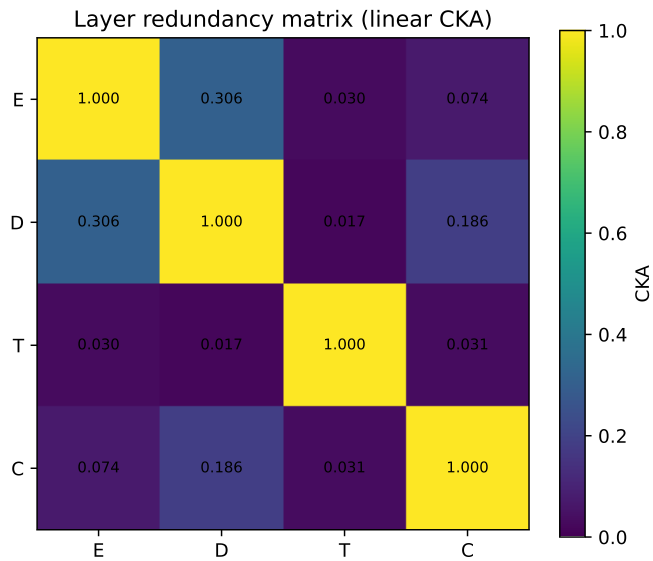

Fig. 7. CKA redundancy matrix. Low-to-moderate values indicate that the feature families are not trivial linear duplicates.

VII. Dataset Provenance, Robustness, and Simulated Control

This section reports the substrate-level evidence that supports the central claim: dataset provenance and quality control, the perturbation robustness of the evidence streams, and the simulated embodied affective-control loop. The aggregate outputs, figures, and tables underlying every statement here are produced by a single reproducible pipeline and are mapped to their source scripts and input fingerprints in the accompanying reproduction map.

A. Dataset Provenance and Quality Control

Table X records the dataset provenance and integrity as verifiable facts: source and access conditions, the included and excluded subjects, the observation and condition counts, the channel and epoch geometry, and the subject-grouped split protocol. Subject 127 is excluded for a recording anomaly, leaving 211 subjects and 633 trial-averaged observations balanced across the three affective conditions. Numerical-validity checks (finite-value fractions, channel and time-shape consistency) and raw-file completeness are summarised in the integrity figure. Subject-level data are maintained in a restricted research environment; the reported outputs are aggregate or hashed.

B. Perturbation Robustness of the Evidence Streams

The evidence streams are characterised under four perturbation families—temporal jitter, amplitude noise, channel dropout, and graph edge perturbation—at the representation level across all configurations, with a bounded raw-signal diagnostic that recomputes a reservoir embedding within fold. The objective is to report the operating regime of each stream rather than a single endpoint, including any boundary at which a reservoir or graph stream degrades faster than the spectral band-power baseline.

TABLE X
Dataset provenance and integrity. Source wording is reused from the verified manuscript data-availability statement. Subject-level data are maintained in a restricted research environment.

Item	Value	Source / verification	Relevance
Data source	SHAPE Community dataset, Laboratory for Clinical Affective Neuroscience, Stony Brook University (access-controlled)	manuscript data-availability (verified)	provenance
Subjects included	211	feature pickle	sample size
Excluded subjects	1 (subject 127, recording anomaly)	Ch5 QC	data quality
Observations	633	feature pickle	evaluation units
Conditions	3 (Negative, Neutral, Pleasant)	feature pickle	affective task
Channels	34	feature pickle	spatial sampling
Downsampled rate	256 Hz	extraction pipeline	temporal sampling
Post-stimulus timepoints	256	feature pickle	epoch length
Preprocessing	baseline-corrected, trial-averaged ERP	extraction pipeline	signal conditioning
Clinical metadata	available (exploratory validation only)	clinical CSV	context labels
Split protocol	subject-grouped cross-validation	analysis code	generalisation
Privacy rule	aggregate/hashed outputs; subject-level data restricted	package policy	data governance

C. Simulated Embodied Affective-Control Loop

We evaluate the substrate as a perceptual estimator inside a simulated closed-loop affective-control task. The expected-free-energy controller is defined explicitly (belief state, likelihood, transition, preferred distribution, risk and ambiguity terms, action-selection rule, and stopping criterion) and compared against passive, random, pragmatic-only, epistemic-only, and perfect-perception oracle policies under action-determined transition noise. The loop is a simulation over recorded observations and is not a physical embodiment.

D. Evaluation Coverage and Reproducibility

Table XI reports the evaluation coverage of the ARSPI-Net components against the system-level evaluation requirements addressed here; it is an internal-consistency summary, not a ranking against other systems. Table XII is the reproducibility panel for the evaluation. Bounded graph observables and a runtime/resource summary supporting the reservoir–graph substrate claim are reported without any hardware-energy measurement.

VIII. Discussion

A. Main Finding

The four ARSPI-Net blocks are operationally distinct under a fixed evaluation protocol. Affective classification is strongest

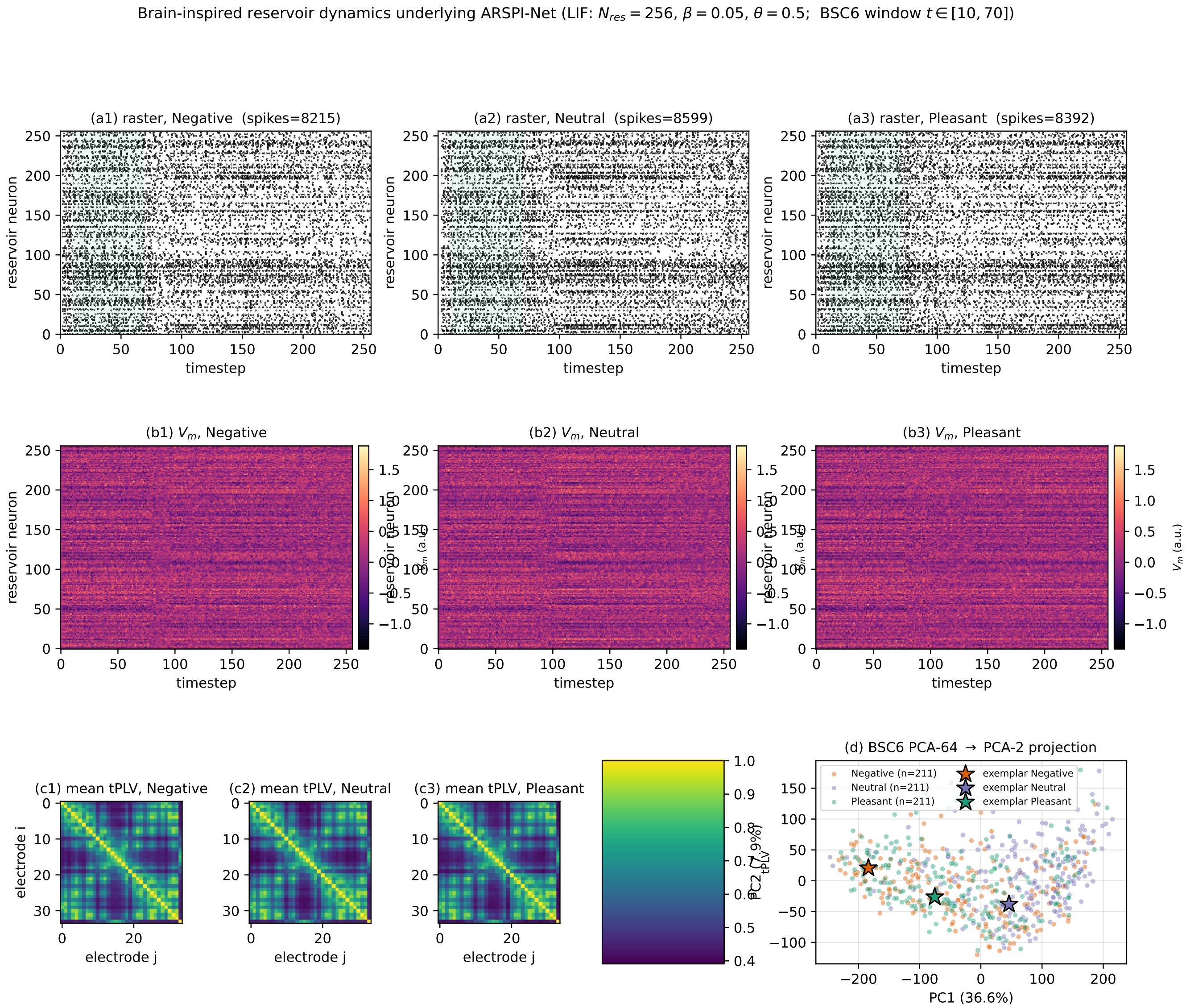

Fig. 8. Reservoir dynamics underlying ARSPI-Net. (a) LIF spike rasters for one exemplar per affective class on a representative channel; the BSC_6 encoding window $t \in [10, 70]$ is shaded. (b) Membrane-potential heatmaps for the same exemplars. (c) Per-class average 34×34 tPLV connectivity matrices across all 211 subjects. (d) BSC_6 PCA projection (PC1 vs PC2) coloured by class with the three exemplars marked.

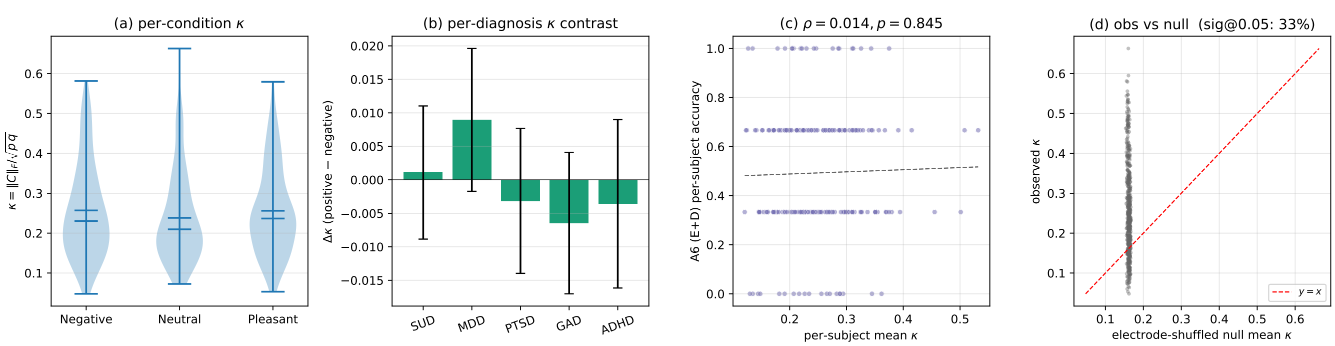

Fig. 9. Structure–function coupling κ. (a) Violin plots of κ per affective condition. (b) Per-diagnosis κ contrast (positive minus negative); none significant under BH-FDR. (c) Per-subject mean κ vs per-subject A6 $(E + D)$ accuracy ($\rho = 0.01$). (d) Observed κ vs the electrode-shuffled null mean; markers above the $y = x$ line have observation-level $p < 0.05$ (33.5% of observations).

for embedding-containing configurations and survives subject-respecting permutation testing for 8 of 10 configurations; standalone T and C do not. Clinical-label sensitivity has a different layer profile in raw means and does not survive

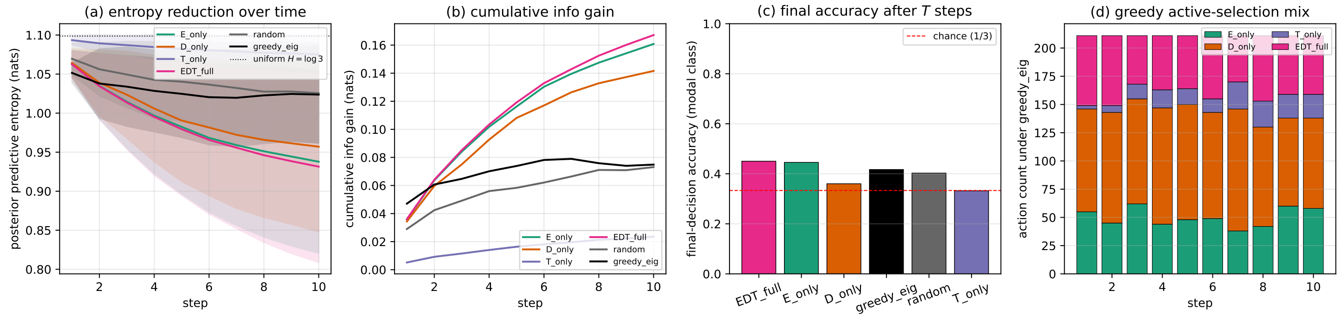

Fig. 10. Sequential evidence accumulation simulation. (a) Posterior predictive entropy vs step under each policy (shaded ± sd across 211 held-out subjects); the dotted line marks the uniform prior $H = \log 3$. (b) Cumulative information gain. (c) Final-decision accuracy (modal class) after $T = 10$ steps; the dashed line marks chance. (d) Action mix selected by the greedy expected-information-gain meta-policy. The loop is closed-loop on the belief side and open-loop on the environment side.

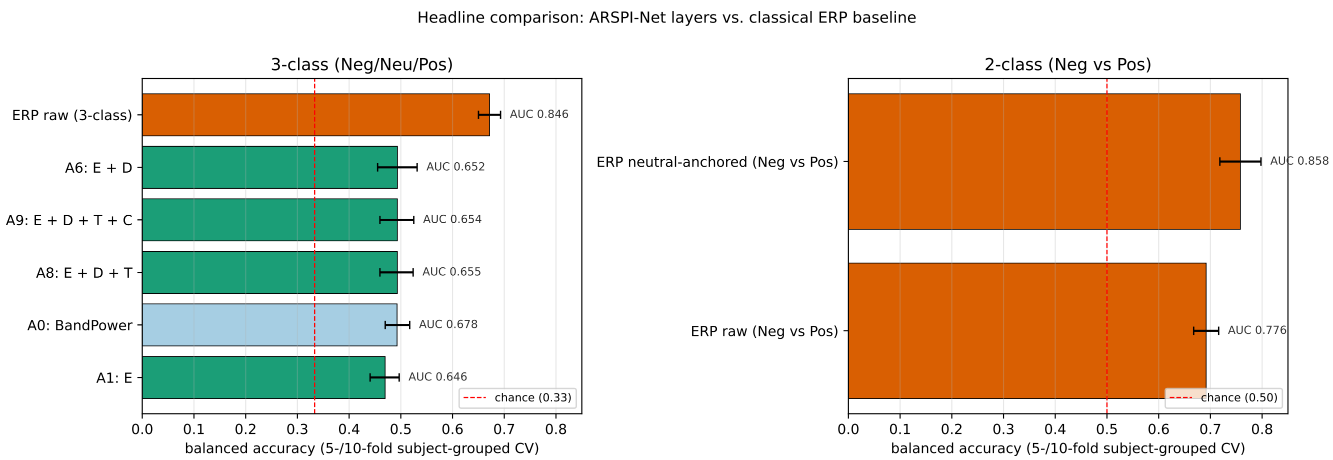

Fig. 11. Headline comparison. ARSPI-Net configurations, the BandPower control, and the classical ERP baseline under matched subject-grouped cross-validation. Numerical AUCs are annotated next to each bar. Chance is marked in red.

TABLE XI
EVALUATION COVERAGE OF ARSPI-NET RELATIVE TO SYSTEM-LEVEL REQUIREMENTS. ROWS ARE ARSPI-NET COMPONENTS; COLUMNS ARE EVALUATION REQUIREMENTS ADDRESSED IN THIS PAPER.

Component	Signal	Mechanism	Perturb.	Ablation	Closed-loop	Runtime
E (reservoir embedding)	✓	✓	✓	✓	✓	✓
D (dynamical)	✓	✓	✓	✓	✓	✓
T (graph topology)	✓	✓	✓	✓	✓	✓
C (coupling)	✓	✓	✓	✓	[illegible]	✓
Closed-loop control	·	✓	✓	·	✓	✓

TABLE XII
REPRODUCIBILITY PANEL FOR THE EVALUATION. SUBJECT-LEVEL DATA ARE RESTRICTED; RELEASED ARTIFACTS ARE AGGREGATE OR HASHED.

Item	Value
Subjects	211 (subject 127 excluded)
Observations	633
Affective classes	3 (Negative, Neutral, Pleasant)
Channels	34
Split protocol	subject-grouped cross-validation
Feature blocks	BandPower, E, D, T, C
Perturbation types	temporal jitter, amplitude noise, channel dropout, graph
Random seeds	42, 43 (profile: pilot)
Closed-loop episodes	≥1000 per policy/ϵ
Policy set	passive, random, pragmatic, epistemic, EFE, oracle
Repo commit	`f18f2d3322`
Upstream commit	`6219ab43ef`
Data availability	restricted; aggregate/deidentified outputs only

BH-FDR across the 30 clinical cells, an outcome the power calculation in Section VI-C predicts for a cohort of this size at these effect magnitudes. Pairwise CKA is low to moderate, and the largest pair, $\mathrm{CKA}(E, D) = 0.31$, is well below the value that would mark linear redundancy. κ rejects an electrode-randomised null in roughly one third of observations and is uncorrelated with classifier accuracy, behaving as a complementary systems-level readout. The embodied agent ranks the four feature-stream policies by per-step expected information gain in the same order as the offline ablation. The blocks behave like different signal-processing views of one EEG response. The analyses support claims about operational layer utility, feature-block validity, FDR-corrected affective inference, structure–function coupling against an electrode-permutation null, and the embodied perception–action ranking. They do not support diagnostic biomarker claims, because no clinical-label cell survived BH-FDR correction.

B. Why This Is Not Merely an Ablation Table

Ablation in many architecture papers is used to prove that every module raises the final number. That is not the result here. Adding C to $E + D + T$ does not move the affective number. T and C alone are weak affective classifiers under permutation inference. Yet the clinical-label matrix has different best-fit layers for different diagnoses, κ rejects its electrode-randomised null in a substantial subset of observations, and the embodied agent's per-step expected-information-gain ranking discriminates the streams. A block can be sub-threshold as a classifier and still operationally distinct as a measurement view, and it can carry usable structure for an embodied agent that an offline endpoint metric cannot see.

C. Brain-Inspired Computing for Embodied AI

The brain-inspired claim rests on named mechanisms rather than analogy. The reservoir is a biophysically reduced LIF network [1], [4]; the embedding is a binned spike-count code with a hardware-native counterpart on event-driven substrates [33]; the topology comes from temporal phase-locking values that have a direct biological reading [23], [24]; and κ measures structure–function coupling of the kind catalogued in network neuroscience [25]–[27]. The embodied claim rests on an implementable sequential-evidence-accumulation protocol whose state update (10) is on-line and requires no end-to-end backpropagation; the loop is closed-loop on the belief side and open-loop on the environment side, and we treat it as a sequential-decision precursor to active inference rather than an active-inference instantiation. The empirical ranking matches the offline ablation: $E + D + T$ is fastest, T-only is slowest. The simulation is a first stage. The next stage is to substitute physiological-feedback actions for our condition-class actions and to run the loop on a neuromorphic accelerator.

D. Implementability on Neuromorphic Hardware

A second reason the ARSPI-Net decomposition is worth taking seriously is that it deploys. The reservoir is fixed, with fewer than $66{,}000$ recurrent weights at $N_{\text{res}} = 256$, and is replicated independently across the 34 EEG channels. Each per-channel reservoir is a 256-neuron LIF population, well below the $8{,}192$-neuron capacity of a Loihi 2 neuro-core [34]; the full $256 \times 34 = 8{,}704$-neuron stack therefore fits inside two of the 128 neuro-cores available on a single Loihi 2 chip. The recurrent matrices are not trained on chip and can be loaded as ROM. SpiNNaker offers an analogous mapping with explicit MAP-and-route software support [35]. The intermediate representation is a binned spike count, which event-driven hardware computes natively in on-chip counters. The PCA compression across 34 channels at 64 components per channel sits at roughly 3.34 million scalar weights and is the dominant trainable parameter count; the linear readout adds a further $2{,}176 \times K$ weights for K output classes. Both projections run host-side on the embedded x86 cores attached to either platform, so re-fitting per subject or per session is a matter of refreshing a small number of weights off the spiking substrate. End-to-end SNN classifiers with millions of synaptic weights on the substrate [9] do not enjoy this property: their entire backbone has to be retrained any time the upstream task changes. A staged neuromorphic architecture in which only the readout adapts is closer to the deployment economics of an embedded system than an end-to-end trainable network is. The reported mappability is a feasibility statement on neuron counts, core counts, and weight counts; a hardware benchmark with measured energy and latency is left for future work.

E. Toward Continual Learning and Synaptic Plasticity

The architecture as reported here is static. The reservoir does not adapt and the readout is fitted offline. The closed-loop simulation is open-loop on the substrate side; the agent's actions modulate which stream it attends to, but no synapse changes. Three concrete extensions preserve the operational-distinctness property while adding biologically motivated adaptation. FORCE learning [5] on the readout weights gives an on-line update rule that is stable in the spectral-radius regime used here. Reward-modulated spike-timing-dependent plasticity [32], [33] on reservoir-to-readout synapses gives a continual-learning rule that the embodied loop's own posterior-entropy reduction can drive as a reward signal. Synaptic homeostasis between sessions, applied to the readout only, gives a simple offline consolidation step that prevents unbounded drift of the linear weights. All three extensions are local, none touches the reservoir core, and each can be evaluated against the same operational-distinctness criteria adopted here.

F. Comparison to Brain-Inspired Embodied-AI Architectures

ARSPI-Net occupies a specific corner of the brain-inspired computing landscape. End-to-end spiking classifiers on EEG [9] embed an SNN inside a single trainable pipeline; their architectural blocks are not separately interpretable and their decision is a single endpoint. Reservoir computing for robotic control couples a fixed dynamical reservoir to a low-dimensional motor readout but does not typically expose intermediate measurement layers. Active-inference work in cognitive robotics [29]–[31] treats the perception module as a generative-model black box rather than as a stack of biophysical primitives. ARSPI-Net combines a fixed spiking reservoir, an event-driven code, an electrode-network topology, and a structure–function coupling readout into one stack, audits the stack as a sequence of operationally distinct measurement layers, and exposes the same stack to a sequential-decision simulation as a set of competing perceptual policies. This is the setting in which staged neuromorphic architectures can be defended as multi-stream measurement substrates for embodied AI, rather than as black-box classifiers.

G. Dataset and Methodological Positioning

Three properties separate this study from neighbouring affective-EEG work. The cohort is large by clinical-affective standards, with 211 community-recruited adults, 633 condition-balanced observations, and five overlapping psychiatric labels; most published affective EEG datasets are smaller and recruited from healthy populations. The architecture is a four-block decomposition of a spiking pipeline rather than an end-to-end classifier, and each block is anchored to a named neural mechanism. The audit protocol treats operational distinctness as a measurable property, with permutation inference, FDR correction, redundancy analysis, an electrode-permutation null on κ, and an embodied closed-loop test. None of these on its own is novel; their combination on this dataset, with claims bounded by the FDR result, is.

H. Clinical Interpretation

Clinical findings are bounded. Balanced accuracies between roughly 51% and 56% are not diagnostic performance, and after BH-FDR across the 30-cell matrix none reach significance ($\min p_{\mathrm{BH}} = 0.30$). Two readings remain plausible. The first is transdiagnostic heterogeneity: psychiatric labels may be expressed through distributed temporal, dynamical, and spatial organisations rather than one universal EEG feature space, in which case no single layer-by-diagnosis cell would cleanly separate. The second is statistical: a sample of 211 community-recruited adults with overlapping comorbidity and observed effect sizes of $\sim 5\%$ over chance is below the power threshold for FDR-corrected detection, and the present analysis fixes the next-step replication target at $N \approx 400$ per diagnosis. Both readings are compatible with the FDR null, and the layer-profile pattern is reported as a directional observation that informs the replication design.

IX. Limitations

The scope of this paper is bounded in five specific ways. The embodied evaluation is a closed-loop *simulation* over recorded EEG observations; it is not physical robot embodiment and not live deployment. The clinical analysis is exploratory validation structure only and is not diagnostic validation; no biomarker claim is made. The runtime and resource figures are wall-clock and structural; no hardware energy is measured and no low-power wearable feasibility is claimed. The dataset is a private, access-controlled clinical resource, which limits public benchmarking and external replication to approved-access conditions. Finally, all quantitative results are restricted to the measured SHAPE ERP regime and should not be read as universal claims about EEG graph neural networks. The remaining paragraphs detail these bounds.

The clinical labels are comorbid and cannot be interpreted as disorder-specific mechanisms from these analyses alone. After BH-FDR correction across the 30 clinical contrasts no test reaches significance, and the layer-by-diagnosis pattern is reported as a weak-signal observation rather than a discovered biomarker. The power calculation in Section VI-C sets the replication target at $N \approx 400$ per diagnosis.

The cohort is cross-sectional and adult. The data span community-recruited participants in a single age band and the analysis is not longitudinal. Architectural maturation, plasticity-driven adaptation, and developmental questions that would otherwise be in scope for a TCDS submission are not addressed by these data and are explicitly excluded from the claims here.

The reservoir is fixed within and across sessions. No synaptic plasticity is implemented. This is the same trade-off that reservoir computing has always made, and it buys interpretability and reproducibility at the cost of adaptivity. The plasticity extensions in Section VIII-E are sketched but not evaluated.

The classical ERP baseline reaches 67.2% balanced accuracy on the three-class affective task and ARSPI-Net reaches roughly 49% under the same readout. The gap reflects the ERP feature set's prior knowledge about affective response components, not a deficit in the brain-inspired stack on its own measurement criteria. The contribution claimed here is the operational-distinctness audit and the embodied test, not endpoint supremacy on the three-class affective task.

The IAPS stimulus set is one of several affective elicitation paradigms. The pipeline has not been evaluated on alternative stimulus modalities (music, video, autobiographical recall), so generalisation outside the IAPS frame is untested.

The embodied demonstration is a simulation, not a robotic implementation. The agent's state model is a Dirichlet posterior over the held-out subject's affective response, the exteroceptive features are EEG-derived, and the action set is a choice over feature-stream policies. A physical-embodiment study would substitute closed-loop sensorimotor feedback for the policy choice, run the loop on a neuromorphic accelerator, and apply continual-learning rules of the kind discussed in Section VIII-E. We treat the simulation as a first step that exposes the layer-utility question in an embodied frame and leave the hardware deployment to future work.

X. Conclusion

We asked whether the four blocks of a staged neuromorphic EEG architecture do different things or just rearrange the same information. Under a fixed evaluation protocol and FDR-corrected permutation inference, eight of ten affective configurations exceed chance and standalone T and C do not. Clinical-label cells differ in raw means but do not survive multiple-comparisons correction across the 30-cell matrix; the present analysis reports them as weak-signal sensitivity. CKA between the blocks is low to moderate. κ rejects an electrode-permuted null in 33.5% of observations and behaves as a complementary systems-level readout uncorrelated with classifier accuracy. A sequential-decision simulation that treats the four streams as competing perceptual policies ranks them in the same order as the offline ablation by per-step expected information gain; the loop is closed-loop on the belief side and open-loop on the environment side, and we position it as a precursor to active inference rather than as one. The blocks are operationally distinct, and the architecture supports the brain-inspired-computing-for-embodied-AI framing on grounds of mechanism (LIF spiking, BSC_6 event-driven coding, tPLV connectivity, κ coupling) and on grounds of an embodied closed-loop demonstration. Clinical claims remain bounded by the FDR result. The methodological proposal is to evaluate staged biomedical architectures not by whether each block raises endpoint accuracy, but by what each block measures, how each block behaves as a perceptual stream available to an embodied agent, and how each block deploys on neuromorphic hardware.

Data and Code Availability

The analysis code, figure-generation scripts, reproduction map, and changelog that produce every quantitative claim in this paper are maintained in the project repository (link provided under approved-access and review conditions). Raw EEG, clinical metadata, and subject-level feature artifacts are maintained in a restricted research environment. The manuscript reports aggregate, deidentified experimental outputs and methodological details sufficient for reproduction under approved data-access conditions. The SHAPE Community dataset is access-controlled and available from the Laboratory for Clinical Affective Neuroscience at Stony Brook University, subject to approval and a data-use agreement. Aggregate, deidentified results are reported; restricted subject-level data are not publicly distributed and require approved access. Committed prediction-level artifacts use hashed subject identifiers.

Acknowledgment

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